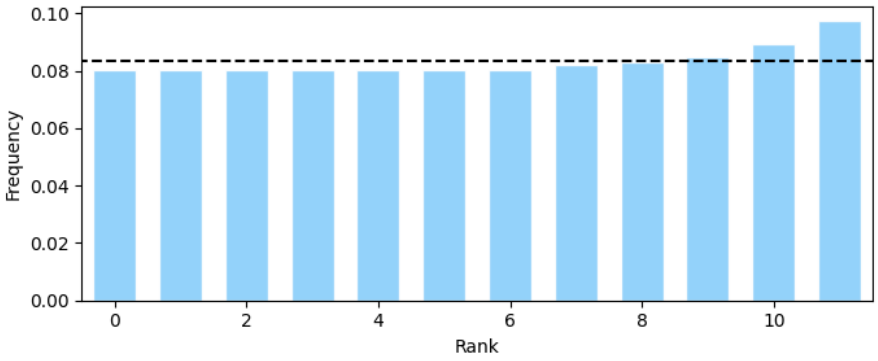

Figure 7. Rank Histogram. Blue bars showing the probability of the observed precipitation's rank among the ensemble members. The black dashed line The black dashed line represents the case where the true value is equally likely to fall between any two ensemble values.

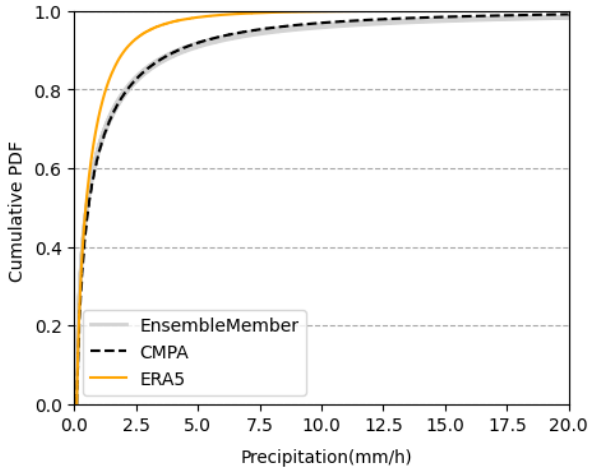

Figure 8. Cumulative PDF of precipitation. The yellow line represents ERA5 precipitation, the black dashed line represents CMPA precipitation, and the gray curves represent the 11 ensemble members' forecasts.

4.3 Real-time ensemble forecasting

We also develop a real-time high-resolution precipitation forecasting system using publicly available ECMWF data as input for our model. Since the ECMWF real-time open dataset provides forecasts at 3-hour intervals, we employ the Pangu-Weather model to perform 2-hour extrapolations, allowing us to obtain hourly atmospheric forecasts. Based on this data, we construct a high-resolution ensemble precipitation forecasting system for the next 120 hours.

To evaluate the system's performance in real-world scenarios, we compare the forecast results against CMPA precipitation as a reference. We assess the Critical Success Index (CSI), Probability of Detection (POD), and False Alarm Ratio (FAR) metrics. The forecast initialization times are at 12:00 UTC for each day from August 1 to August 31, 2021, with a forecast range of 120 hours.

The CSI scores exhibit a periodic fluctuation with a 3-hour cycle, likely due to the inference predictions introduced by the Pangu model. Overall, for forecast lead times within 24 hours, for a threshold of 0.1 mm/h, the average CSI score across the 11 ensemble members ranges between 0.2 and 0.35, while for 2 mm/h, the CSI score is approximately 0.1. As expected, higher thresholds lead to lower CSI scores, reflecting the increasing difficulty in accurately predicting heavy precipitation events (this may be attributed to the relative rarity of high-intensity rainfall events, which makes them more challenging to predict).

As the forecast lead time increases, the FAR gradually rises, and the CSI shows a decreasing trend. However, even by lead time of 5 days, the CSI for the 0.1 mm/h threshold remains at a reasonable level of 0.15–0.3, while for 2 mm/h, it stays around 0.05. We also plot the CSI scores for the best-performing ensemble member at each forecast time. These results indicate that with the adoption of a more advanced post-processing or ensemble integration method, forecast accuracy could be further improved by optimally selecting or weighting the best ensemble members.

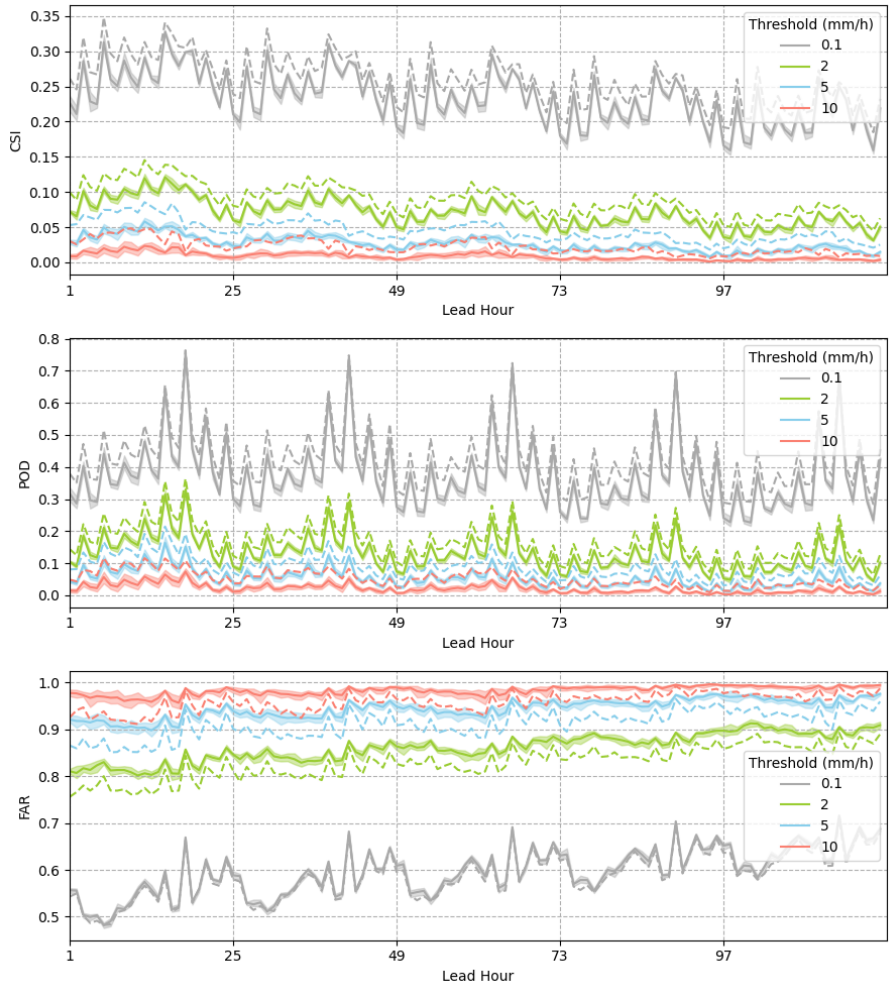

Figure 9. The average CSI, POD, and FAR from the real-time forecast system for August 2021, over 120-hour forecasts issued at 12 UTC each day. Shaded areas show the range of values across individual ensemble members; the solid line shows average value across all members; the dashed line indicates the metrics of the best-performing member.

5 Conclusion

This study proposes a physics-inspired deep learning framework that combines deterministic and probabilistic modeling approaches to significantly improve the accuracy and spatial resolution of high-resolution precipitation forecasting. The model is trained using ERA5 and CMPA datasets, with atmospheric variables such as temperature, geopotential, humidity, and wind as input, and produces high-resolution precipitation forecasts at 0.05° resolution. Built on a 3D Swin Transformer backbone, the deterministic model significantly improves diagnostic accuracy, especially for moderate to heavy rainfall, through improvements in patch embedding/recovery, loss function, and the incorporation of geographic and temporal features. With atmospheric state and meso-scale precipitation as a constraint, a conditional diffusion model in latent space is applied to capture uncertainties at convective scales after compressing the data using VAE. During inference, ensemble forecasts are generated by random sampling of Gaussian noise in the latent space. Our evaluation shows that the constructed ensemble forecast system is nearly unbiased, and its precipitation distribution is closer to CMPA compared to ERA5. Additionally, we developed a real-time forecast system based on ECMWF's public dataset, which demonstrated robust performance in terms of CSI scores.

Overall, this study presents a novel approach to high-resolution precipitation forecasting. Due to data limitations, when transferring the model trained on ERA5 to ECMWF data, we didn't apply fine-tuning beyond adjusting the precipitation data scale in this study. Despite the high homogeneity between the two datasets, differences in resolution and other factors still introduce some discrepancies. In the future, as more data becomes available, we will apply model fine-tuning to further enhance the real-time forecast system's performance.

Data Availability

The ERA5 dataset used in this study was accessed from the Google Cloud public data archive, which can be found at https://console.cloud.google.com/storage/browser/gcp-public-data-arco-era5. The ECWWF open real-time forecast with 0.25° resolution is available at https://console.cloud.google.com/marketplace/product/bigquery-public-data/open-data-ecmwf; The CMPA precipitation dataset was provided by the China Meteorological Administration's National Meteorological Information Center.

Code Availability

Acknowledgments

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